Imperative Interference: Does Social Register Shape Instruction Topology in Large Language Models?

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Note on version 3. The headline result of version 2, that declarative rewriting of one instruction block reduces cross-linguistic variance by 81%, did not replicate, and we withdraw it as a claim about how to write prompts. Two things went wrong, and either alone would have been enough. First, the declarative variant changed three things at once (it changed the mood, dropped the intensifiers in favour of a uniform list, and added an explicit "during commit operations" scope that the original lacked), so the 81% could never have been credited to register alone. Second, every March response in this paper was generated with three trials at temperature 0, and we have since learned two things about that regime: the cells are not reproducible (the headline cell gave a different answer in October on what OpenRouter's records identify as the same model snapshot, served by the same provider), and the `llm_judge` scorer, used for 18 of the 22 probes (version 2 said seven), was called at the provider's default temperature, so nothing guaranteed deterministic scores even when the responses repeated. In October 2026 we re-ran the comparison on current models, with the confound split out (a 2×2 of mood by sentence shape) and with ten sampled trials per cell at temperature 0.7 (§4.5). The declarative rewrite had no detectable effect on cross-linguistic spread or on adherence. The one effect that survived sampling, an exploratory contrast on one block, runs the other way from our design advice: rewriting the block as one scoped imperative per sentence *lowered* adherence, and the same sentences in declarative mood recovered it. The σ figures (5.9σ, 11.1σ, 3.5σ) are withdrawn; the rank $p = 0.045$ stands only as a description of the March 2026 snapshot. The topology results (§4.2, §4.6) were measured in the same regime and have not been re-run; they should be read as snapshot observations, and the social-register mechanism as a hypothesis, not an established effect. Version 3 also corrects four errors of description that version 2 carried: the number of judge-scored probes, the account of Gemini's E-PROC cells, the text of the declarative variant, and the code availability statement. The text of version 2 is otherwise retained, with the affected claims marked. The title, which stated the mechanism as a finding, is now a question.

Note on version 2. This version corrects two reported statistics to match the analysis script of record. The control-probe mean shift is +0.0010 (was reported as +0.0013 in v1) and the declarative-rewrite effect is 5.9σ above the 21 control probes (was 5.8σ); the permutation p-value is 0.045. The reconciliation was a bookkeeping correction between the manuscript and the audit script, not a change of analysis. The central finding—declarative rewriting reduces cross-linguistic variance, with zero of 21 control probes exceeding the observed effect—is unchanged.

Abstract

System prompt instructions that cooperate in English compete in Spanish, with the same semantic content, but opposite interaction topology. We present instruction-level ablation experiments across four languages and four models, and the hypothesis (stated in version 2

as a finding) that this topology inversion is mediated by *social register*: the imperative mood carries different obligatory force across speech communities, and models trained on multilingual data have learned these conventions. All of these measurements are March 2026 snapshots taken at generation temperature 0 with three trials. In the March 2026 snapshot, declarative rewriting of a single instruction block reduced cross-linguistic variance by 81% ($p = 0.045$, rank-permutation vs. 21 control probes); *this did not replicate* in an October 2026 re-run with ten sampled trials per cell, which also separated mood from sentence shape (version 3 note, §4.5). The surviving register result is narrower and exploratory: when this one block was rewritten as one scoped instruction per sentence, imperative mood lowered adherence and declarative mood recovered it. In the snapshot, rewriting three of eleven imperative blocks shifted Spanish instruction topology from competitive to cooperative, with spillover effects on unrewritten blocks; this has not been re-run under sampling. These observations suggest that models may process instructions as social acts, not technical specifications: "NEVER do X" is an exercise of authority whose force may be language-dependent, while "X: disabled" is a factual description. If register mediates instruction-following at inference time, it plausibly does so during training. We state this as a testable prediction: constitutional AI principles authored in imperative mood may create language-dependent alignment. Corpus: 22 hand-authored probes against a production system prompt decomposed into 56 blocks. Total experimental cost: $69 USD (verified against OpenRouter billing).

1 Introduction

System prompts for LLM-based agents are written as commands. "NEVER use TodoWrite during commits." "ALWAYS prefer dedicated tools over Bash." "Use the Task tool VERY frequently." These instructions work in English. They fail in predictable, structurally characterizable ways when the same instructions are translated to other languages.

In our March 2026 measurements, the failure was not uniform degradation but structural inversion. In English, the instructions in a production system prompt form a cooperative network: removing any instruction reduces overall adherence. In Spanish, the same instructions, which are semantically identical and correctly translated, form a competitive network: removing some instructions *improves* adherence. The instructions are interfering with each other.

Prior work on multilingual prompting establishes that prompt language affects LLM performance [Zhang et al., 2025, Mondshine et al., 2025, Yin et al., 2024]. These studies measure main effects: aggregate performance drops when prompts are translated. We measure something different: the *interaction structure* between instructions. The distinction matters. A main effect tells you that translation hurts. An interaction topology tells you *how*: which instructions interfere, which cooperate, and whether the pattern is stable across languages.

(Version 3: the next two paragraphs state the mechanism as version 2 held it. The single-block evidence for it did not replicate, and the topology evidence has not been re-measured, so read them as the hypothesis, §4.5.) Our central finding was that this topology difference is mediated by **social register**. Imperative mood ("Use X," "NEVER do Y") carries different obligatory force in different speech communities. In English, stacked imperatives create a coherent authority context. In Spanish, the same stack creates competing obligation signals. Declarative register ("X: enabled," "Y: disabled") sidesteps the social dimension entirely by stating facts rather than issuing commands.

This is not a parsing problem. It is a sociolinguistic one. Models trained on multilingual data have internalized different conventions for how authority and obligation are encoded across languages. When a system prompt exercises authority through imperative mood, the model's response depends on which language's conventions it applies.

We investigated this through a five-experiment arc (all March 2026 snapshots):

1. **Observation:** Instruction interaction topology is a three-way interaction (model × language × instruction), not a language main effect (§4.1).

2. **Falsification:** Information density does not explain the topology differences (§4.3).
3. **Single-block fix (withdrawn in v3):** In the March snapshot, declarative rewriting reduced cross-linguistic variance for one instruction by 81%, $p = 0.045$ (rank-permutation vs. 21 control probes; §4.4); it did not replicate under sampling (§4.5).
4. **Topology confirmation:** Pairwise ablation (Haiku, temperature 0) showed the cooperative/competitive inversion between English and Spanish (§4.2).
5. **Causal mechanism (unreplicated):** In the March snapshot, rewriting three imperative blocks to declarative shifted Spanish topology from competitive to cooperative, with spillover to unrewritten blocks (§4.6); Haiku only, not re-run.

The experimental substrate is the Claude Code system prompt (v2.1.50), decomposed into 56 classified blocks in prior work [Mason, 2026]. That analysis identified 21 static interference patterns through structural evaluation and multi-model scouring. The present work asks: do those instructions *behave* the same way when translated?

Our contributions are:

- The first instruction-level ablation study of a production system prompt across languages (4 models × 4 languages × 22 probes).
- An observation (March snapshot, Haiku, not re-run under sampling) that instruction interaction topology inverts across languages: cooperative in English, competitive in Spanish.
- A proposal that social register is the mechanism. (Version 3: the single-block confirmation did not replicate; what survives is a mood effect on adherence within one controlled sentence shape, §4.5.)
- An observation of spillover (same caveats): rewriting the register of some instructions changed adherence on others.
- A testable prediction: if register mediates instruction-following at inference time, alignment training via imperatively-phrased constitutional principles may be language-dependent.

2 Background and Related Work

2.1 Multilingual Prompting

The multilingual prompting literature has grown rapidly, but overwhelmingly measures main effects. Zhang et al. [2025] study cross-lingual system prompt steerability across five languages and find significant performance variation, but treat the prompt as monolithic with no instruction-level decomposition and no interaction measurement. Mondshine et al. [2025] compare translation strategies (translate vs. keep in target language) and find that translation direction matters. Wang et al. [2025] show that linguistic cues activate different cultural knowledge, establishing that language is not just a vehicle for content. Yin et al. [2024] measure how politeness interacts with language and culture, finding that politeness affects LLM outputs, which is the closest work to our register finding, but measuring only main effects, not interaction topology.

Our gap relative to all of this work: nobody measures pairwise instruction interactions across languages, nobody has identified interaction topology as language-dependent, and nobody has proposed register as the mechanism.

2.2 Speech Acts and Register

The observation that utterances *do* things (not just say things) originates with Austin and Searle's speech act theory. "NEVER use TodoWrite" is not a description; it is an exercise of authority. The distinction between locutionary content (what is said), illocutionary force (what is done), and perlocutionary effect (what results) offers a way to read our March observations: identical locutionary content produced different perlocutionary effects across languages, which

would follow if the illocutionary force of imperative mood is language-dependent. (Version 3: this is the hypothesis, not a demonstrated cause.)

Register is the sociolinguistic term for the variety of language used in a particular social context. Imperative register ("Do X") invokes authority. Declarative register ("X is the case") states facts. The choice of register is a social act: it positions the speaker relative to the listener. When a system prompt uses imperative register, it positions itself as an authority issuing commands. When it uses declarative register, it positions itself as a knowledge source stating properties.

Geng et al. [2026] study authority claims as a framing mechanism in LLM instructions, finding systematic shifts in instruction prioritization. This is the closest work to ours in framing, but they do not test cross-linguistic effects or measure interaction topology.

2.3 System Prompt Analysis

Mason [2026] present Arbiter, a framework for detecting interference in system prompts through structural evaluation and multi-model scouring. Applied to three vendor prompts (Claude Code, Codex CLI, Gemini CLI), they identify 152 scourer findings and 21 hand-labeled interference patterns. Their analysis is static: it identifies structural contradictions but does not measure runtime behavior. The present work uses the same corpus (Claude Code v2.1.50, 56 blocks) and asks whether the structural patterns manifest differently at runtime across languages.

3 Methodology

3.1 Corpus

Our experimental corpus is the Claude Code v2.1.50 system prompt, decomposed into 56 contiguous blocks classified by tier (system/domain/application), category (identity, security, tool usage, workflow, etc.), modality (mandate, prohibition, guidance, information), and scope [Mason, 2026]. Of 56 blocks, 22 are *free* (ablatable) as their presence can be toggled without breaking tool definitions or security policy. The remaining 34 are *constrained*: removing them would cause tool-calling failures or security violations.

3.2 Translation

The 56-block corpus was translated to Mandarin Chinese (zh), French (fr), and Spanish (es) using Gemini Flash 2.0 (`google/gemini-2.0-flash-001`) via OpenRouter. We used a non-Anthropic model to avoid circularity (Claude translating its own instructions).

Translation rules preserved markdown formatting, kept tool names and API identifiers untranslated, maintained imperative tone, and translated structural markers (e.g., "IMPORTANT:" was translated to the equivalent marker in each target language).

Table 1: Corpus size after translation.

Language	Characters	% of English
English	15,970	100%
Mandarin	6,910	43.3%
Spanish	19,178	120.1%
French	20,744	129.9%

Mandarin is 57% shorter by character count. This size difference is a confound we address in §4.3.

3.3 Models

Table 2: Models tested via OpenRouter.

Model	OpenRouter ID	Training Bias
Claude Haiku 4.5	`anthropic/claude-haiku-4-5`	English-heavy
Gemini Flash 2.0	`google/gemini-2.0-flash-001`	English-heavy
DeepSeek V3	`deepseek/deepseek-chat-v3-0324`	Chinese + diverse
Mistral Med. 3.1	`mistralai/mistral-medium-3.1`	French + diverse

The model selection is deliberate: two English-primary models (Haiku, Gemini), one Chinese-primary (DeepSeek), one French-primary (Mistral). If language effects were purely about English-centrism in training, all four should degrade similarly. They do not.

3.4 Probe Battery

Twenty-two hand-authored probes test adherence to each free block. Each probe consists of a user message designed to elicit the target behavior, a scoring method, and expected/violation descriptions. Scoring methods include:

- `not_contains`: checks absence of prohibited patterns (e.g., emoji characters)
- `length`: scores inversely with response length against a baseline
- `llm_judge`: the same model evaluates its own output against judge criteria

All probes are in English: the user speaks English, only the system prompt changes language. Three trials per probe at temperature 0.0. (Version 3: that temperature applied to generation only. The judge call set no temperature, so judging ran at each provider's default.)

The `llm_judge` method (18 of 22 probes; version 2 said seven) uses the same model being tested. This is a known limitation (§5.4).

3.5 Ablation Design

We use covering arrays of strength 2 to generate ablation configurations. A strength-2 covering array guarantees that every pair of blocks appears in at least one configuration where both are present and one where one is absent, enabling measurement of pairwise interactions without exhaustive enumeration.

Phase 0 (single-block removal): each of 22 free blocks is removed individually while all others remain present. This measures main effect: how much each block contributes to overall adherence.

Phase 1 (pairwise): a strength-2 covering array generates configurations where pairs of blocks are co-absent, enabling measurement of pairwise interaction effects beyond additive main effects.

3.6 Statistical Methods

Welch's t-test with exact t-distribution CDF (not normal approximation) for small samples ($n = 3$ trials). **Benjamini-Hochberg** FDR correction for multiple comparisons. **Permutation tests** (100,000 permutations) for non-parametric significance testing of variance reduction and hub concentration.

3.7 Experimental Sequence

Total API cost across all experiments: $68.95 USD, verified against OpenRouter billing records (Table 4).

Table 3: Experimental arc: each experiment's result motivated the next.

Label	Experiment	Finding	Motivated
T7	Cross-ling. baseline	Three-way interaction	T9
T9	E-DENSE (padding)	Density is bidirectional	T10
T10	E-PROC (rewrite)	Declarative fixed variance (did not replicate)	E-PAIR-ES
—	E-PAIR-ES (pairwise)	Spanish topology inverted	T11
T11	E-TOPO (topology)	Consistent with register (not re-run)	—

Table 4: Verified API costs from OpenRouter billing (March 20–22, 2026). Haiku dominates cost because it serves as both subject and judge for 18 of 22 LLM-judged probes, doubling its call volume.

Model	Calls	Cost (USD)	% of Total
Claude Haiku 4.5	14,270	$66.10	95.9%
Mistral Med. 3.1	1,560	$1.21	1.8%
DeepSeek V3	1,560	$1.12	1.6%
Gemini Flash 2.0	1,728	$0.53	0.8%
Total	19,118	**$68.95**	

4 Results

4.1 Cross-Linguistic Baselines

Table 5: Mean adherence scores across all 22 probes (baselines, all blocks present).

Model	English	Mandarin	French	Spanish	Range
Haiku	**0.853**	0.783	0.774	0.736	0.117
Gemini	**0.781**	0.765	0.693	0.755	0.088
DeepSeek	**0.799**	0.786	0.771	0.757	0.042
Mistral	0.726	**0.819**	0.753	0.788	0.093

Three of four models perform best in English (Table 5, Figure 1). Mistral is the anomaly: it performs worst in English (0.726) and best in Mandarin (0.819). A French-trained model peaks in neither its training language nor English. DeepSeek is the most language-robust (range 0.042); Haiku is the most sensitive (range 0.117).

At the aggregate level, this looks like a language main effect with a training-bias interaction. But aggregates conceal the structure. At the probe level, the phenomenon is a **three-way interaction: model × language × instruction.**

Probe-level inversions. On `commit-restrictions` (should separate commit workflow from TodoWrite usage), Haiku scores 1.00 in English and 0.00 in Mandarin. Gemini shows the *exact opposite*: 0.00 in English, 1.00 in Mandarin. Same instruction, same translation, opposite behavioral effect per model.

On `explore-agent` (should delegate to Explore agent for complex analysis), Haiku collapses from 1.00 in English to 0.22 in Spanish. The word "Explore" functions as both a tool name and a common verb; Spanish translation preserves the semantic meaning ("explorar") but loses the proper-noun binding.

On `use-task-for-search`, Haiku *improves* from 0.50 in English to 1.00 in Mandarin. The Mandarin translation added bold emphasis to "prefer," resolving an ambiguity present in the

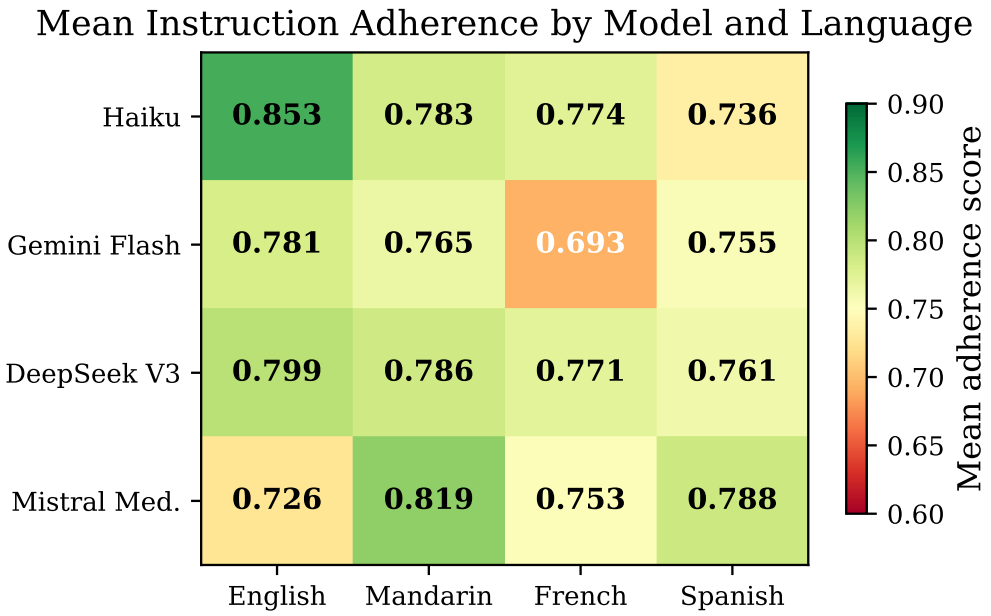


Figure 1: Cross-linguistic baseline adherence. Three models perform best in English; Mistral performs *worst* in English and best in Mandarin; a French-trained model peaking in neither its training language nor English.

English original.

No single factor predicts the outcome. You cannot say "Spanish degrades instruction following" as it degrades Haiku (−13.8%) but improves Mistral (+8.5%). You cannot say "this instruction is robust" because `commit-restrictions` is perfect for Haiku in English and zero in Mandarin. The interaction terms are larger than the main effects.

4.2 Topology Inversion

In the March snapshot (temperature 0, three trials, not re-run), Phase 0 single-block removal on Haiku indicated that instruction topology is language-dependent (Figure 2).

Table 6: Phase 0 main effects (Haiku): mean Δ when each block is removed. Negative = cooperative (removal hurts). Positive = competitive (removal helps). Selected blocks shown.

Block	English	Mandarin	French	Spanish
no-time-estimates	−0.140	−0.094	−0.021	+0.078
no-new-files	−0.099	+0.064	+0.002	+0.066
no-overengineering	−0.098	+0.074	−0.020	+0.033
todowrite-repeated	−0.090	+0.009	−0.045	+0.057
objectivity	−0.054	+0.015	−0.032	+0.102

English: All main effects are negative. Removing any block hurts overall adherence. The instructions form a cooperative network: every block contributes.

Spanish: Most main effects are positive. Removing blocks *improves* adherence. The topology inverts from cooperative to competitive. The translated instructions interfere with each other.

French: Mixed, mostly weak. No clear hub structure. Flat topology.

Mandarin: Hub partially preserved (`no-time-estimates` remains strongest at −0.094) but with positive effects elsewhere. Mixed topology.

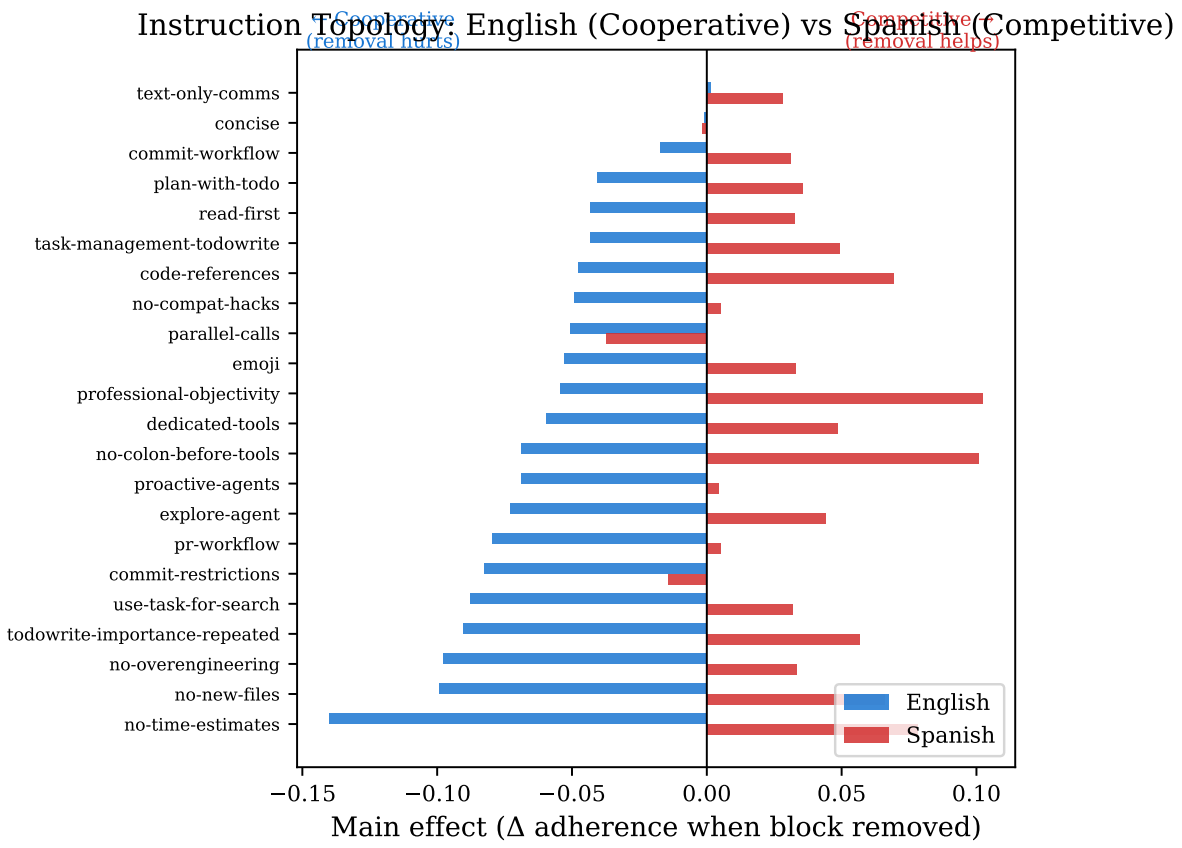


Figure 2: Instruction topology comparison (Haiku, Phase 0 main effects). English effects are uniformly negative (cooperative); Spanish effects are predominantly positive (competitive). The same instructions that strengthen the English prompt weaken the Spanish one.

In the March snapshot, the cross-linguistic correlation of main effects was consistent with the inversion: English and Spanish are anti-correlated ($r = -0.274$). No language pair has a positive correlation.

Pairwise confirmation (E-PAIR-ES). Phase 1 pairwise ablation on Haiku with the Spanish corpus showed the same topology at the interaction level (a temperature-0 snapshot, not re-run):

- English mean pairwise Δ: −0.116 (cooperative)
- Spanish mean pairwise Δ: +0.010 (competitive)

Hub significance was confirmed via permutation test (100,000 permutations): `no-time-estimates` appears in 15 of 20 top English interactions ($p < 0.00001$). The hub concentration is real, but it is language-specific. The same block is not the hub in every language.

4.3 Falsification: Information Density

The Mandarin corpus is 57% shorter than English by character count (Table 1). Could the topology differences be driven by information density rather than language?

E-DENSE tested this by padding the Mandarin corpus with semantically neutral filler to match English character count. Prediction: padded Mandarin would show higher cross-model variance (lower agreement), because compression was driving convergence.

Result: hypothesis falsified. Overall cross-model variance *decreased* slightly with padding (0.1189 → 0.1041). The aggregate effect is opposite to the prediction. But the aggregate conceals a bidirectional mechanism:

- **Compression causes mode confusion:** On `text-only-comms`, Gemini switched from tool-code output to natural language when the prompt was padded (0.0 → 1.0). The compressed instruction was too terse for Gemini to distinguish "communicate in text" from "use tools."
- **Compression aids procedural focus:** On `explore-agent`, Haiku crashed from 1.0 to 0.28 with padding. The dense prompt overwhelmed its delegation logic.

Information density operates through two opposing mechanisms: compression aids procedural focus but causes mode confusion. The net effect depends on the model×instruction pair. Density is a confound, not a mechanism. The topology differences between languages are not explained by length alone.

4.4 Declarative Rewriting Reduces Variance

If the topology differences are not about density, what explains them? The encoding taxonomy (§4.7) shows that procedurally encoded instructions have 2.9× the cross-linguistic variance of declaratively encoded ones. **E-PROC** tested whether rewriting a procedural instruction to declarative form would reduce its cross-linguistic variance.

The `commit-restrictions` block was rewritten in two forms:

- **Declarative:** a single scope sentence ("During commit operations, the following tools and commands are unavailable") followed by a flat list with per-item status ("TodoWrite tool: disabled"). (Version 2 quoted text here that does not appear in the variant; this is the text that was run.)
- **Scoped:** bracketed block with compact inline list

Both preserve the same semantic content. All other blocks unchanged (21 control probes). (Version 3: the original block is an "Important notes:" header over bullets with intensifiers ("NEVER", "IMPORTANT", "ALWAYS") and no statement that the rules apply during commits. The declarative variant changed the mood, replaced the intensified bullets with a uniform status list, and added that scope in its opening sentence. The comparison below therefore cannot attribute its effect to mood; §4.5 separates mood from sentence shape.)

Table 7: E-PROC: cross-linguistic variance of `commit-restrictions` by encoding variant. Mean variance across 4 models.

Variant	Mean Variance	Reduction
Original (procedural)	0.1567	—
Declarative	0.0290	81% ($p = 0.045$)
Scoped	0.0966	38% (n.s.)

In the March snapshot, the declarative variant reduced cross-linguistic variance by 81% ($p = 0.045$, rank-permutation vs. 21 control probes). Version 2 reported this as 5.9σ above the 21 control probes (mean control $\Delta = +0.0010$). Version 3 withdraws that figure. It divides by the spread of the control probes' changes, and that spread was measured in a regime whose dependence structure we do not know: responses at temperature 0, scored by a judge at the provider's default temperature, three times each. Whether it understates or overstates the noise, it is not a calibrated estimate of it. With zero of 21 control probes exceeding the observed effect, the rank result was margin-thin even on its own terms: a single control probe above the effect would cross $p > 0.05$. The effect is carried by Haiku: restricted to Haiku alone the reduction is 100%, and excluding Gemini it is 77%. (Version 2 gave these as 11.1σ and 3.5σ; they use the same construction as the 5.9σ and are withdrawn with it.) Variances in Table 7 are sample variances of the four language means; the rank test and the replication use population variances, which changes the values but not the ranks. The most dramatic change: Haiku Mandarin went

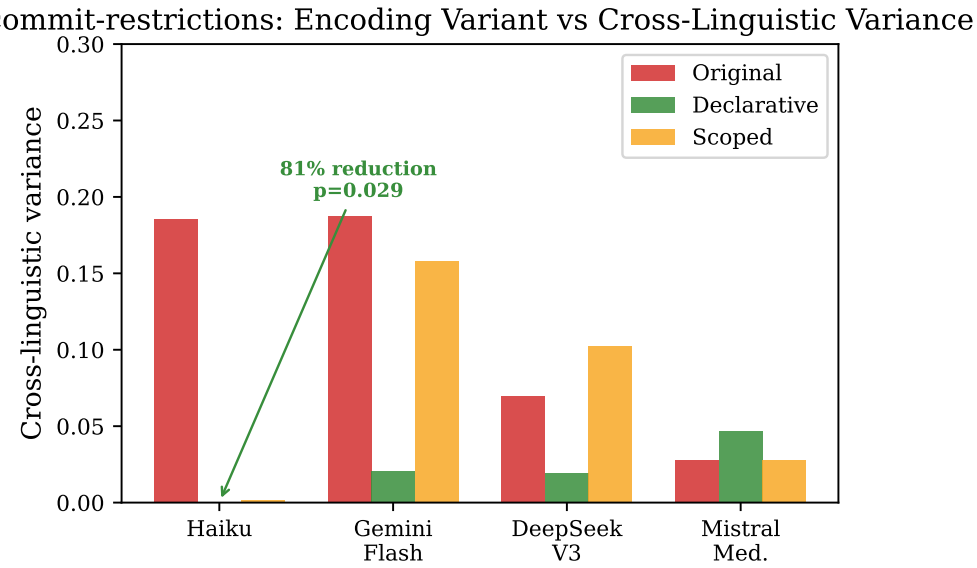


Figure 3: Cross-linguistic variance of `commit-restrictions` by encoding variant and model (March 2026 snapshot). Declarative rewriting eliminates Haiku's cross-linguistic variance almost entirely. (Version 2's caption said Gemini was unaffected; its variance also fell, see text.)

from 0.00 to 1.00, a cell that in October scored 1.00 under the original block as well.

Version 2 said Gemini was unaffected, scoring 0.00 in every language and variant. That was wrong. Gemini's original-block cells were $[0, 0, 0]$ in English, French and Spanish and $[1, 1, 1]$ in Mandarin; its declarative cells were $[0, 0, 0]$ except Spanish, $[0.5, 0.5, 0]$. Its cross-linguistic variance therefore fell too (population variance 0.1875 to 0.0208), by moving its one passing language to failure rather than by fixing anything, which is a reminder that a variance reduction is not the same thing as an improvement.

4.5 Replication Under Sampling (Version 3, October 2026)

The E-PROC result rests on three trials per cell, with responses generated at temperature 0 and, for 18 of the 22 probes including the target, scored by an LLM judge called at the provider's default temperature. Across the 16 March cross-linguistic baseline runs, 257 of the 351 complete cells (73%) returned the identical score on all three trials and 111 (32%) the identical response text; only 91 were identical in both, and 20 cells with identical text received different scores. These counts do not establish the dependence between trials or the effective sample size, and we cannot say how much information three trials carry. The cell that carried the headline (Haiku, Mandarin, original block: $[0, 0, 0]$, becoming $[1, 1, 1]$ when rewritten) is one in which all three trials agreed.

We re-ran the comparison twice in October 2026, with predictions sealed and timestamped before each run. Both runs used a 2×2 design on the `commit-restrictions` block, crossing mood (imperative or declarative) with sentence shape (the original loose bullets, or a controlled shape with one instruction per sentence, the commit scope stated in each instruction sentence ("When you commit, ..."; a few supporting sentences, such as "Git commands are available," carry no scope), no intensifiers, and at most twenty words per sentence). The loose declarative cell is the version 2 declarative text, unchanged. Everything else in the prompt, the probe battery and the scoring was as in E-PROC.

The first re-run kept the March regime (three trials, temperature 0). On the three models that passed a response-validity gate, mean cross-linguistic variance on the target rose from 0.0133 (original) to 0.0207 (declarative), and the reason is in the cells: current Haiku no longer

fails the original block in Mandarin, so there was nothing for the rewrite to fix. The model alias had changed since March (`claude-haiku-4-5` is no longer served), but OpenRouter's per-call activity records show that the March E-PROC calls and the October calls were served by the same snapshot (`claude-4.5-haiku-20251001`) through the same provider (Amazon Bedrock). Under the same recorded snapshot, provider and generation temperature, then, the headline cell did not reproduce. Those labels cannot rule out a change in how the provider serves the model, so the cause remains open; what the episode shows is that a temperature-0 cell is not a stable measurement over months. The controlled-shape imperative cell looked nine times worse than the original, and two controlled-shape cells (one imperative, one declarative) collapsed to zero; in a follow-up on Haiku at temperature 0.7 with ten trials, those collapses recovered, and low-scoring responses appeared in many cells rather than in a few.

The second re-run was the one designed to answer the question: four models, four languages, all four variants, the target probe and six controls, ten trials per probe at temperature 0.7, and a trial-level bootstrap (10,000 draws for the target, 2,000 for the controls). The results are in Table 8.

Table 8: Sampled replication (October 2026). Differences pooled over four models (Haiku 4.5, DeepSeek V3, Mistral Medium 3.1, Gemini 3.7 Flash); 95% bootstrap intervals. Spread is the variance of the four language means on the target probe; level is mean adherence. orig. = original block; decl. = version 2 declarative text; ctrl. = controlled sentence shape; imp. = imperative.

Comparison	Change in spread	Change in level
orig. → decl. (E-PROC)	−0.0003 [−0.014, +0.016]	−0.036 [−0.097, +0.026]
orig. → ctrl. imp.	+0.0121 [−0.003, +0.032]	**−0.072** [−0.136, −0.007]
ctrl. imp. → ctrl. decl.	−0.0108 [−0.032, +0.008]	**+0.113** [+0.048, +0.177]

The declarative rewrite of version 2 had no detectable effect on spread or on level. That is a statement about detection, not equivalence: the level interval admits anything from a loss of about ten points to a gain of about three, and the spread interval is wide enough to include a reduction as large as the March one, measured against today's much smaller baseline spread. Of the six sealed predictions, two held, and the three that bet *for* the register account all failed, including the one that predicted a spread reduction within the controlled shape. What survived is an exploratory contrast (P5 sealed a level gain for the loose declarative rewrite, which failed; no sealed prediction concerned this within-shape adherence contrast), and it runs in the opposite direction from our design advice: rewriting the block as one scoped imperative per sentence lowered adherence (for Haiku alone, from 0.864 to 0.615), and the same sentences in declarative mood recovered it (Haiku 0.878; pooled, not detectably different from the original, +0.041 [−0.018, +0.099]). The intervals are unadjusted for the several contrasts we report. Mood therefore mattered for adherence within that one controlled shape of one block; we have no evidence that it reduces cross-linguistic variance, and none that it improves on the original block. Six control probes, drawn at random from the 21 with the seed fixed before the run, were checked for spillover; none met the sealed criterion for a change in spread (an interval lying entirely above 0.01). That criterion does not establish that the controls were unchanged: one control's interval reaches 0.024, and in an exploratory check one control's adherence fell by about three points between the two controlled variants.

Two limits apply to the replication itself. It covers one block of one prompt, and the judge for several probes is the model under test, as in the original study. Gemini 3.7 Flash answered many control probes with tool-call attempts and no text (519 recorded failures, almost all on control probes; 95.6% of its target-probe trials were valid), and it scored near ceiling on the target in every cell (its spreads were between 0 and 0.0025) and, pooled with equal weight, lowers every variant's pooled spread (by 0.0025 to 0.0068 depending on the variant). Raw responses,

judge outputs, recorded failures, preregistrations and analysis scripts are in the repository (`docs/research/RESULT_e_ste.md`, `docs/research/RESULT_e_ste_sampled.md`).

4.6 Register Shapes Topology

In the March snapshot, E-PROC appeared to show that declarative rewriting fixes individual instruction variance (it did not replicate, §4.5). **E-TOPO** tested whether it also fixes the topology inversion.

Three blocks showing the strongest competitive effects in E-PAIR-ES were rewritten from imperative to declarative register in the Spanish corpus:

- `proactive-agents`: "Debes usar proactivamente..." → "Herramienta Task: Estado: disponible..."
- `use-task-for-search`: "prefiere usar la herramienta Task..." → "Preferencia de herramienta: Task..."
- `todowrite`: "Utiliza estas herramientas MUY frecuentemente..." → "Frecuencia de uso: muy alta..."

Phase 1 pairwise ablation was repeated with the rewritten corpus.

Table 9: E-TOPO: topology shift after declarative rewriting (Spanish, Haiku).

Condition	Mean Δ	Direction	Competitive probes
Original (imperative)	+0.010	Competitive	7/22
Rewritten (declarative)	−0.055	Cooperative	4/22

Target probes (rewritten blocks) shifted as expected:

Table 10: E-TOPO: target probes (rewritten blocks).

Probe	Orig. Δ	Decl. Δ	Shift	Result
proactive-agents	+0.274	−0.380	−0.655	Fixed
todowrite	+0.155	−0.023	−0.177	Fixed
use-task-for-search	+0.118	+0.047	−0.071	Reduced

`proactive-agents` shows the largest shift in the entire dataset: from the most competitive probe (+0.274) to one of the most cooperative (−0.380).

Spillover effects. Three *unrewritten* blocks also shifted from competitive to cooperative:

Table 11: E-TOPO: spillover effects (unrewritten blocks).

Probe	Orig. Δ	Decl. Δ	Shift	Result
no-compat-hacks	+0.123	−0.267	−0.389	Fixed
plan-with-todo	+0.012	−0.174	−0.186	Fixed
todowrite-repeated	+0.011	−0.059	−0.070	Fixed

(Version 3: E-TOPO is a Haiku-only, three-trial, temperature-0 measurement, taken in the same regime in which the E-PROC headline cell proved fragile. It has not been re-run under sampling, so the paragraph below states the hypothesis as we held it, not an established effect.) In version 2 we wrote that the spillover is the strongest evidence for the social register hypothesis: that the imperative register creates system-wide interference, not just per-block fragility. Reducing the number of imperatives from approximately eleven to eight (by rewriting

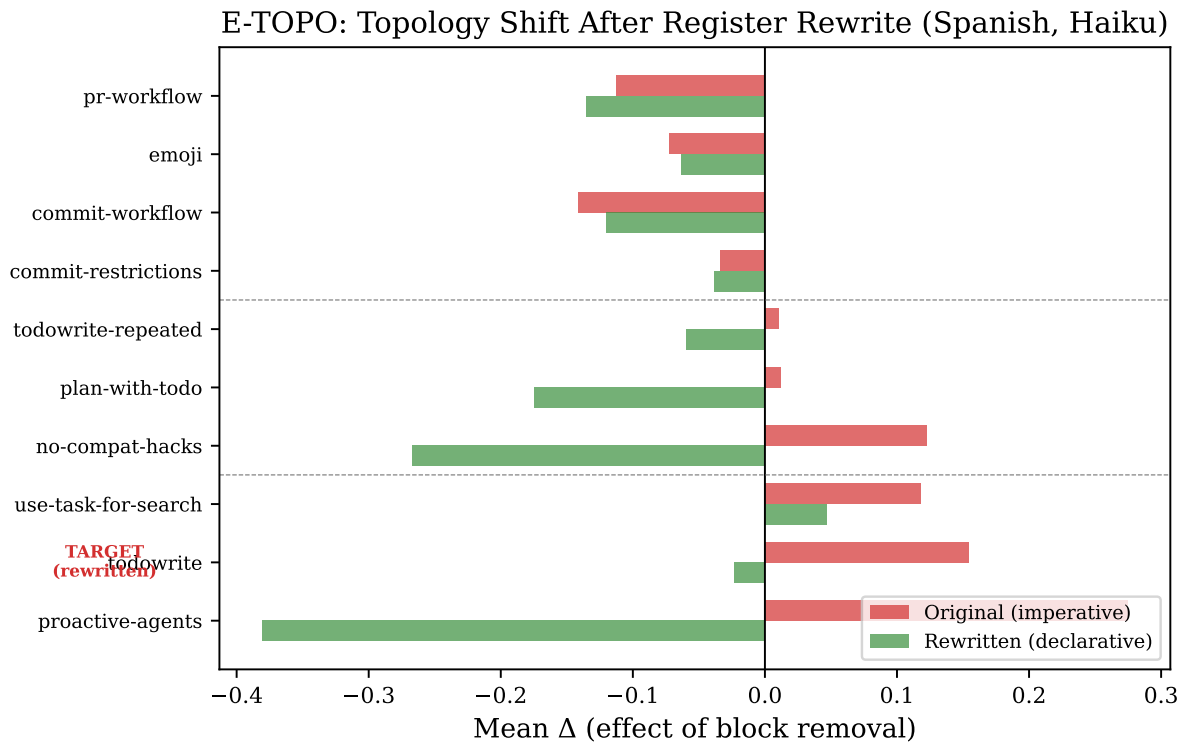


Figure 4: E-TOPO topology shift. Target probes (top) shift from competitive to cooperative as expected. Spillover probes (middle) also shift despite being unrewritten. Control probes (bottom) remain stable. In version 2 we read the spillover as showing that register operates at the system level, not per-block; it is an unreplicated snapshot.

three) reduces the total obligation-resolution load, freeing the model to process the remaining blocks more accurately.

Control probes that were already cooperative remained cooperative (`commit-restrictions`: $-0.033 \rightarrow -0.038$; `commit-workflow`: $-0.141 \rightarrow -0.120$). The rewrite did not destabilize existing cooperative interactions.

4.7 Encoding Taxonomy

Classifying the 22 free blocks by encoding style reveals a systematic fragility pattern:

Table 12: Cross-linguistic variance by encoding style.

Encoding	Blocks	Mean Variance	Example
Procedural	11	0.0514	"When X, do Y not Z"
Declarative	11	0.0175	"Do X" / "Don't do X"
Ratio		**2.9**×	

Procedural instructions—those describing conditional workflows ("during commits, don't do X, also don't do Y")—are 2.9× more fragile under translation than declarative instructions that state rules directly. The fragility has two mechanisms:

Procedural compression. Conditional workflow chains compress ambiguously under translation. The model loses track of which constraints apply to which context. (Version 2 called this fixable with declarative rewriting, citing E-PROC; that confirmation did not replicate, §4.5.)

Domain jargon opacity. Technical terms that are clear in English developer culture don't translate cleanly. `no-compat-hacks` ("backwards-compatibility hacks") has variance 0.1132, the highest among declarative blocks. Mistral scores 0.00 in English and French but 1.00 in Mandarin and Spanish. The French-trained model paradoxically fails in its native language.

5 Discussion

5.1 Instructions as Social Acts

The specification model of system prompts, treating instructions as formal constraints on output, predicts language-invariant interaction topology. If "NEVER use TodoWrite" is a constraint, removing it should have the same structural effect regardless of what language it's expressed in. Our March data contradict this prediction. The same instruction is essential in English and deadweight in Spanish.

The social act model, which we offer as a hypothesis, predicts what we observed: that interaction topology depends on how the target language encodes the social relationship between instructor and instructed. Stacked imperatives in English reinforce a single authority frame. In languages where imperative mood is more socially loaded, more personal and more direct, the same stack creates competing obligation signals, as if from multiple authority sources. The model must resolve not just "what am I being told to do" but "who is telling me, and do these authorities agree?"

We took the spillover effect (§4.6) to be the strongest evidence for this interpretation; as noted there, it is an unreplicated temperature-0 observation. If register operated per-block, rewriting three blocks should affect only three probes. Instead, unrewritten blocks also shifted from competitive to cooperative. This is consistent with a system-level obligation-resolution load: each imperative consumes processing resources, and reducing the count benefits the whole prompt.

On this hypothesis, declarative register sidesteps the social dimension: "Status: available" does not invoke authority, so it cannot compete for it. (The sampled replication found no detectable benefit from the declarative rewrite of the original block.)

5.2 Alignment Implications

We state the following as a **testable prediction**, not a confirmed result: if social register mediates instruction-following at inference time (which version 2 claimed our experiments demonstrate, and which the sampled replication supports only within one controlled sentence shape of one block, as an exploratory effect on adherence), then it plausibly mediates instruction-following during training as well.

Constitutional AI [Bai et al., 2022] trains models using principles that are overwhelmingly written in imperative mood: "Be helpful." "Be harmless." "Be honest." These are social acts exercising authority. If imperative register carries different obligatory force across languages, then:

1. Models trained on imperatively-phrased principles may develop language-dependent alignment: helpful in English, differently-helpful in other languages, not because of the *content* of the principles but because of their *register*.
2. The Mistral anomaly is consistent with this. A French-trained model that performs worst in English and best in Mandarin suggests that training language creates behavioral signatures that interact unpredictably with register.
3. The direction of the interaction is not predictable from the language pair alone. Our March data show that English-Spanish produced topology inversion, English-French topology flattening, and English-Mandarin partial preservation. These are different failure modes, not different magnitudes of the same failure.

This prediction is testable. One could compare models trained on declaratively-phrased constitutional principles against models trained on imperatively-phrased ones, measuring cross-linguistic alignment consistency. The register hypothesis predicts that declaratively-trained models would show more consistent alignment across languages.

The broader implication: the question "why aren't Spanish-speaking countries building Spanish-primary models?" is not merely economic. Our March data suggest that training language may create behavioral signatures that interact with instruction register in ways that are only beginning to be characterized.

5.3 Design Principles

Version 3. The first two rules below are *not supported* by the sampled replication (§4.5). Declarative rewriting of the original block had no detectable effect, and the controlled shape that rule 2 describes (one self-contained, scoped instruction per sentence) *lowered* adherence when written in the imperative. What the replication supports is narrower than a rule: for this one block, rewriting it as one scoped imperative per sentence cost adherence and the declarative version of the same sentences did not, an exploratory result on one block of one prompt that we would want replicated before anyone treated it as advice. Rule 3 was never directly tested. We leave the version 2 list in place so that the change is visible.

The experimental findings (version 2) suggested these design rules for system prompts intended to work across languages:

1. **Declare facts, don't issue commands.** "X: disabled" transfers more reliably than "NEVER use X." Factual descriptions sidestep the social dimension of register.
2. **Self-contained constraints.** Each rule should be interpretable without context from surrounding rules. Conditional chains ("during X, don't do Y") compress ambiguously under translation.
3. **Examples over concepts** for domain-specific patterns. Code examples survive translation better than abstract concepts like "backwards-compatibility hacks."

These principles are mechanical transformations, not semantic changes. They can be applied to any system prompt without altering its intended behavior.

5.4 Limitations

Single corpus. All results are from one system prompt (Claude Code v2.1.50). We cannot generalize to other prompts without replication.

Machine translation. Gemini Flash translated the corpus. We did not verify translation quality with native speakers. Translation artifacts could explain some findings.

LLM-as-judge. Eighteen of 22 probes (version 2 said seven) use the same model being tested as judge, at the provider's default temperature. This could create systematic bias. Cross-model judging would be more independent.

Limited model set. Four models. The three-way interaction may look different with additional models.

English-only user messages. We tested "system prompt in language X, user speaks English." The case where both system prompt and user speak the same non-English language is untested.

Haiku-only for pairwise and topology. E-PAIR-ES and E-TOPO used only Haiku. The topology inversion may or may not generalize to other models.

Manual register rewrite. The imperative-to-declarative rewriting was done by hand. We did not test automated rewriting.

No bootstrap on topology difference. We observe that English is cooperative and Spanish is competitive, but we have not performed a bootstrap or permutation test comparing effect distributions across languages.

Small sample sizes. Three trials per probe. Per-probe results have high variance.

Temperature 0 (added in version 3). This is the limitation that mattered most, and version 2 did not list it. At temperature 0 the three trials of a cell usually repeat one draw (73% of baseline cells scored identically on every trial), so "three trials" is closer to one observation, and the variance across controls understates the noise it was used to measure. The one result we re-ran under sampling did not survive (§4.5). Every other number in this paper was measured in the same regime and carries the same caveat. A related variable we did not control is the serving provider: OpenRouter's records show that in a single hour of the March E-PROC run, DeepSeek V3 calls were served by eight different providers, any of which may differ in quantization or inference stack. (Haiku calls went almost entirely to one provider, in March and in October alike.)

5.5 Future Work

The experimental arc suggests several research directions, ordered by proximity to current findings:

Phase transition mapping. If the cooperative-to-competitive shift is real, it may not be gradual. Varying imperative density from 0 to 11 blocks could reveal a critical threshold where the topology undergoes a discontinuity. E-TOPO's result (changing 3 of 11 blocks flipped the topology) hints that Spanish/Haiku may be near the critical point.

Instruction ecology. The cooperative/competitive framing maps naturally to ecological models. Lotka-Volterra competition models take species traits and carrying capacity as inputs. If "instruction traits" (register, length, specificity) and "attention carrying capacity" (model × language) can be measured, the ecological model could predict which instruction pairs will compete without running experiments.

Register head. If models process imperative and declarative registers differently, as T11 suggested in the March snapshot, this should be visible in the transformer's internals. Mechanistic interpretability (activation patching, probing classifiers) could identify the attention head(s) that encode register. If found, one could predict register sensitivity from model architecture without behavioral experiments.

Pragmatic force translation. Current machine translation preserves semantic content but not social force. A register-aware translator would map obligatory force to each language's conventions, not just meaning. Whether such a tool is needed depends on the register effect replicating, which the single-block result did not.

Constitutional prompt design. If models process instructions as social contracts, a system prompt could be designed as a formal constitution with explicit precedence rules and a conflict resolution clause. Whether a model told "you are governed by this constitution" would develop more consistent cross-linguistic behavior than one given a flat list of imperatives is an open question with practical implications for alignment.

6 Conclusion

(Version 3.) In version 2 this section said that the register of an instruction determines both its adherence and the interaction topology of the prompt, that the problem is fixable by declaring facts instead of issuing commands, and that the fix spills over to unrewritten instructions. What remains after a sampled replication is smaller. The single-block fix did not replicate on October 2026 models, and its design confounded mood with shape and scope. The topology observations (cooperative in English, competitive in Spanish, flat in French) and the spillover have not been re-measured outside the temperature-0 regime that produced the failed result. The register effect that does survive is a cost of the imperative mood on adherence when a block is written as one

scoped instruction per sentence, and the declarative mood recovers it. That is an exploratory result on one block, and a much narrower claim than the one we made.

The hypothesis we proposed (version 2 text follows) remains a hypothesis. Models trained on multilingual data have learned that imperative mood carries different obligatory force in different speech communities. When a system prompt stacks imperatives, the resulting interaction topology depends on which language's conventions the model applies. We note that "English" and "Spanish" are proxies here for register-encoding patterns, not language-intrinsic properties—varieties within a language (formal vs. colloquial, regional differences in authority encoding) likely produce different topologies for the same reason.

The inference-time finding has a training-time implication. Constitutional AI principles are written in imperative mood. If register mediates instruction-following at inference time, it plausibly does so during training. Alignment may be language-dependent at the register level. This is a testable prediction. We hope someone tests it.

Total cost of the experimental arc that produced these findings: sixty-nine dollars, verified against billing records. Ninety-six percent of that was one model serving as its own judge. The two October replications cost about thirty-four dollars more: $19.31 for the temperature-0 re-run and its follow-up, and $14.36 for the sampled run, summed from OpenRouter's per-call records for the four models within each run's recorded time window.

7 Code and Data Availability

Code, analysis scripts, data files, and reproducibility artifacts are available at `https://github.com/fsgeek/arbiter` (paper snapshot: tag `p3-v3`). Versions 1 and 2 cited tag `v0.2.0` and two scripts, `run_e_proc.py` and `run_e_topo.py`; the tag predates every script used in this paper and the two scripts never existed under those names. The commands below are the ones that produced the results.

```
# March 2026 snapshot (temperature 0, three trials)
python scripts/run_cross_linguistic.py --compare
python scripts/run_fragility_experiments.py --compare proc   # E-PROC
python scripts/run_topology_rewrite_test.py                  # E-TOPO
python scripts/audit_eproc_permutation.py                    # v2 statistics
# October 2026 replication (version 3)
python scripts/run_e_ste.py                  # T=0, all four models
python scripts/analyze_e_ste.py
python scripts/run_e_ste_temp.py             # follow-up: Haiku, target, T=0.7
for m in haiku gemini deepseek mistral; do   # sampled run, 10 trials, T=0.7
  python scripts/run_e_ste_temp.py --model $m --probes sampled \
      --experiment e-ste-sampled
done
python scripts/analyze_e_ste_sampled.py
```

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